

Lightweight and Accurate Channel Prediction Using Mamba in High-Mobility Wireless Networks

Abstract—In high-mobility environments, precise channel state information is essential. Yet, it becomes more challenging due to practical hardware constraints. This challenge often inspires us to conduct studies with cutting-edge models to redefine seamless connectivity. Recently, Mamba, a novel sequence modeling architecture, combines content-based reasoning with efficient linear computational complexity. This paper attempts to explore Mamba’s potential against established temporal networks for predicting fast-fading channels under the 3GPP CDL-A standard. First, we leverage pilot signals to simulate noisy channel observations by generating realistic datasets that incorporate Rayleigh fading and Doppler effects to analyze prediction accuracy, training efficiency, and robustness to channel aging across varying horizons. Afterwards, Mamba’s state-space processing is adjusted for channel prediction. Results show that Mamba consistently outperforms baselines, achieving up to 10 dB lower error and 15% faster convergence than GRU, with strong resilience in both noise-free and noisy scenarios. These findings highlight Mamba’s potential as a lightweight and accurate solution for real-time channel prediction in dynamic wireless networks.

Index Terms—Wireless Communication, Channel Prediction, Channel State Information, Mamba, GRU, CNN, 3GPP, CDL-A, Doppler shift.

I. INTRODUCTION

The evolution toward sixth-generation (6G) wireless networks demands unprecedented accuracy in channel state information (CSI) acquisition to support emerging applications such as extended reality, autonomous vehicles, and industrial IoT [1], [2]. High-mobility scenarios present particular challenges because rapid channel variations and pilot overhead constraints can degrade the performance of conventional estimation techniques, including least squares and minimum mean square error methods [3], [4]. Deep learning (DL) has shown considerable promise in addressing these limitations [1], [5], [6]; however, existing approaches often struggle to maintain stability in long-horizon predictions and to generalize effectively across diverse mobility patterns [7], [8].

Traditional channel prediction methods based on autoregressive models or Kalman filters show noticeable performance gaps in realistic 3GPP channel conditions [3]. DL-based alternatives such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have improved temporal modeling capabilities, but they encounter computational bottlenecks and suffer from error accumulation over extended prediction horizons [9], [10]. More recent approaches incorporate attention mechanisms [10] or physics-inspired architectures [11] to enhance robustness; however, their complexity can grow prohibitively with large antenna arrays and wideband configurations [8], [12]. Generative models offer another direction

[13], but their inference latency is still too high for real-time deployment.

State space models (SSMs) have recently emerged as a promising alternative, combining the ability to capture long-range dependencies with linear computational complexity. The Mamba architecture [14] advances this paradigm through selective state transitions, achieving high efficiency in both language and time-series tasks. Despite these advantages, its potential for complex-valued MIMO channel prediction under standardized 3GPP conditions has yet to be explored. Previous DL studies have tended to focus on narrow scenario settings [15], exhibit limited noise resilience [16], or depend on auxiliary information such as environmental images [8]. In addition, specialized techniques designed for IRS-assisted systems [17] or XL-MIMO deployments [18] often lack generalizability to conventional cellular environments.

This work addresses these gaps by adapting the Mamba architecture to high-mobility channel prediction within the 3GPP CDL-A model. The main contributions are as follows:

- We propose a novel Mamba-based architecture for MIMO CSI prediction that combines selective state updates with input-dependent dynamics, enabling superior long-horizon forecasting compared with conventional RNN and CNN baselines.
- Through experiments on CDL-A channel models with realistic mobility and noise conditions, we show that Mamba achieves significant improvement in normalized mean square error (NMSE) over GRU and CNN baselines for long-horizon prediction, while retaining computational efficiency.
- We provide a detailed analysis of Mamba’s robustness to signal-to-noise ratio variations and channel Doppler spreads, demonstrating its suitability for mobility-intensive 6G scenarios such as high-speed rail and unmanned aerial vehicle communications.

The remainder of this paper is organized as follows. Section II describes the detailed methodology, including CDL-A channel simulation, OFDM pilot signal generation, input-label pair construction, and the proposed neural network architecture. Section III presents the experimental setup, while Section IV discusses the evaluation results. Finally, Section V concludes the paper and outlines potential directions for future research.

TABLE I: NR CDL-A Channel Parameters

Parameter	Value	Description
Carrier Frequency	2 GHz	Operating frequency
Subcarrier Spacing	15 kHz	OFDM subcarrier spacing
Number of Subcarriers	624	Total subcarriers
Time Slots	1000	Simulation duration
Tx/Rx Antennas	4x4	ULA configuration
UE Velocity	20 km/h	Mobility parameter
Doppler Shift	37 Hz	$f_d = \frac{v \cdot f_c}{c}$
Delay Profile	CDL-A	Non-line-of-sight scenario
Number of Taps	23	Multipath components
Cluster Configuration	7 clusters	23 subpaths/cluster

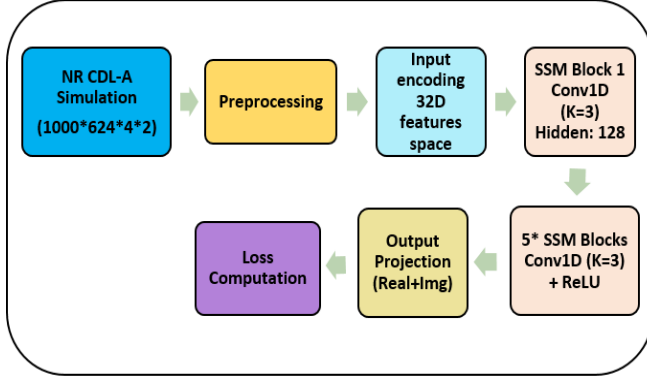


Fig. 1: Proposed Mamba-based neural network architecture for high-mobility MIMO channel prediction.

II. DETAILED METHODOLOGY

A. CDL-A Channel Simulation

The channel modeling relies on the 3GPP CDL-A model to emulate non-line-of-sight (NLoS) urban macrocell propagation. The channel impulse response (CIR) is defined by a sum over clusters and subpaths:

$$H(t, \tau) = \sum_{c=1}^{N_{\text{clusters}}} \sum_{s=1}^{N_{\text{subpaths}}} \alpha_{c,s}(t) e^{j\phi_{c,s}(t)} \delta(\tau - \tau_{c,s}), \quad (1)$$

Here, $\alpha_{c,s}(t)$ represents the complex gain of the s th subpath within the c th cluster, and the Doppler phase shift is defined as:

$$\phi_{c,s}(t) = 2\pi f_d t \cos \theta_{c,s}, \quad (2)$$

with the Doppler shift given by

$$f_d = \frac{v \cdot f_c}{c} \cos \theta_{c,s}, \quad (3)$$

where v is the UE velocity, $f_c = 3.5$ GHz is the carrier frequency, and c denotes the speed of light.

B. OFDM Pilot Signal Generation

Pilot symbols modulated via QPSK are organized into a matrix $X_{\text{pilot}} \in \mathbb{C}^{N_{sc} \times N_t}$ with $N_{sc} = 64$ subcarriers and $N_t = 4$ transmit antennas. The OFDM signal is generated using the inverse discrete Fourier transform (IDFT):

$$\mathbf{x}_{\text{OFDM}} = \mathbf{F}^H X_{\text{pilot}}, \quad (4)$$

Algorithm 1 Mamba State-Space Processing for Channel Prediction

Require: $\mathbf{H}_{\text{noisy}} \in \mathbb{C}^{624 \times 4 \times 4 \times T}$ $\triangleright$ Noisy channel tensor
Ensure: $\hat{\mathbf{H}}[t+p] \in \mathbb{C}^{624 \times 4 \times 4}$ $\triangleright$ Predicted channel

- 1: **Input Encoding**
- 2: $\mathbf{X} \leftarrow \text{concat}(\Re(\mathbf{H}_{\text{noisy}}), \Im(\mathbf{H}_{\text{noisy}}))$ $\triangleright \mathbb{R}^{B \times 5 \times 32}$
- 3: $\mathbf{X} \leftarrow \text{LayerNorm}(\mathbf{X})$ $\triangleright$ Normalize per antenna-pair
- 4: **Selective State-Space Processing**
- 5: **for** each Mamba block $m = 1$ to 6 **do**
- 6: $\Delta \leftarrow \text{Softplus}(\mathbf{W}_{\Delta} \mathbf{X})$ $\triangleright$ Time step parameterization
- 7: $\mathbf{B}, \mathbf{C} \leftarrow \text{LinearProj}(\mathbf{X})$ $\triangleright$ Input-dependent SSM parameters
- 8: $\mathbf{A} \leftarrow \exp(\Delta \otimes \mathbf{A}_{\text{base}})$ $\triangleright \mathbf{A}_{\text{base}} \in \mathbb{R}^{16}$ learned
- 9: $\tilde{\mathbf{B}} \leftarrow \Delta \mathbf{B}$ $\triangleright$ Discretized input matrix
- 10: $\mathbf{Y} \leftarrow \text{ParallelScan}(\mathbf{A}, \tilde{\mathbf{B}}, \mathbf{C}, \mathbf{X})$ $\triangleright$ Linear-time SSM computation
- 11: $\mathbf{X} \leftarrow \mathbf{Y} + \text{DepthwiseConv1D}(\mathbf{X})$ $\triangleright$ Residual connection
- 12: **end for**
- 13: **Output Projection**
- 14: $\hat{\mathbf{Y}} \leftarrow \text{Linear}(\mathbf{X}[:, -1, :])$ $\triangleright$ Last timestep features
- 15: $\hat{\mathbf{H}} \leftarrow \text{ReconstructComplex}(\hat{\mathbf{Y}})$ $\triangleright$ Combine real/imaginary
- 16: **Loss Calculation**
- 17: $\mathcal{L}_{\text{dB}} \leftarrow 10 \log_{10} \left(\frac{1}{B} \sum_{i=1}^B \|\hat{\mathbf{H}}_i - \mathbf{H}_{\text{clean},i}\|_F^2 \right)$

where $\mathbf{F}$ is the 64×64 DFT matrix. A cyclic prefix (CP) of length 16 is appended to mitigate inter-symbol interference.

C. Noise Injection and Received Signal

The simulated received time-domain signal is given by:

$$y[n] = \sum_{k=0}^{K-1} H_k[n] x[n-k] + w[n], \quad (5)$$

where $H_k[n] \in \mathbb{C}^{N_r \times N_t}$ is the k th tap of the channel response, $x[n]$ is the transmitted pilot symbol, and $w[n] \sim \mathcal{CN}(0, \sigma_w^2 \mathbf{I})$ models the additive white Gaussian noise (AWGN). The noise variance is related to the signal-to-noise ratio (SNR) γ (in dB) by

$$\sigma_w^2 = \frac{\|y[n]\|^2}{10^{\gamma/10}}. \quad (6)$$

D. Input-Label Pair Generation

To prepare a dataset for channel prediction, historical noisy channel state information (CSI) is mapped to future (ground-truth) channels. For each sample i , an input-label pair is constructed with a lookback window $L = 5$ and a prediction horizon p . The input is defined as:

$$\mathbf{X}_i = \left[\Re\{H_{\text{noisy}}[t-L]\}, \Im\{H_{\text{noisy}}[t-L]\}, \dots, \Re\{H_{\text{noisy}}[t]\}, \Im\{H_{\text{noisy}}[t]\} \right] \in \mathbb{R}^{L \times N_r N_t},$$

and the corresponding ground-truth is:

$$\mathbf{Y}_i = [\Re\{H_{\text{true}}[t+p]\}, \Im\{H_{\text{true}}[t+p]\}] \in \mathbb{R}^{2 \times N_r N_t}. \quad (7)$$

The NR CDL-A channel model parameters used in the simulation are summarized in Table I. The MATLAB 5G toolbox is used to simulate the dataset and align subcarriers and antennas. Furthermore, the data are normalized (zero mean and unit variance) to ensure stable training dynamics.

E. Proposed Mamba-Based State-Space Channel Prediction Framework

The proposed neural network architecture, illustrated in Fig. 1, is built upon the Mamba framework, which leverages state-space modeling for efficient temporal sequence processing. By integrating input-dependent parameterization and parallel scan operations, the model captures both long-range channel dynamics and local spatial correlations. The following subsections describe the key components and mechanisms enabling robust channel prediction.

Algorithm 1 provides a procedural realization of the proposed Mamba-based channel prediction framework, serving as a bridge between classical state-space modeling and its neural implementation. Starting from noisy CSI observations, the algorithm applies input-dependent discretized state-space transformations to efficiently capture time-varying channel dynamics, while maintaining linear computational complexity. These operations are implemented through stacked selective state-space blocks and parallel scan mechanisms, whose theoretical foundations are introduced next by revisiting state-space modeling and its extension to the Mamba architecture.

1) From State Space Modeling to Mamba Architecture:

For continuous-time, State Space Modeling (SSM) maps input $x(t)$ to output $y(t)$ through latent state $h(t)$ using $\mathbf{A}$, evolution matrix and projection matrices $\mathbf{B}, \mathbf{C}$ as follows:

$$\begin{cases} h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) = \mathbf{C}h(t) \end{cases} \quad (8)$$

To model time-invariant (LTI) system behavior using SSM, we fixed the evolution and projection matrices. However, such attempts fail to capture input-dependent, time-varying scenarios, for instance dynamic channel conditions due to mobility and multipath propagation.

Mamba incorporates the required time-varying behavior into the SSM system through a selection mechanism, which in turn is implemented via discretization and scanning operations as follows

Discretization: Using zero-order hold with step size Δ :

$$\begin{aligned} \tilde{\mathbf{A}} &= \exp(\Delta \mathbf{A}) \\ \tilde{\mathbf{B}} &= (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I})\Delta \mathbf{B} \end{aligned}$$

Scanning: The discretized SSM computes outputs through a parallel associative scan:

$$y_t = \mathbf{C} \left(\prod_{i=1}^t \tilde{\mathbf{A}}_i \right) h_0 + \sum_{k=1}^t \mathbf{C} \left(\prod_{i=k+1}^t \tilde{\mathbf{A}}_i \right) \tilde{\mathbf{B}}_k x_k \quad (9)$$

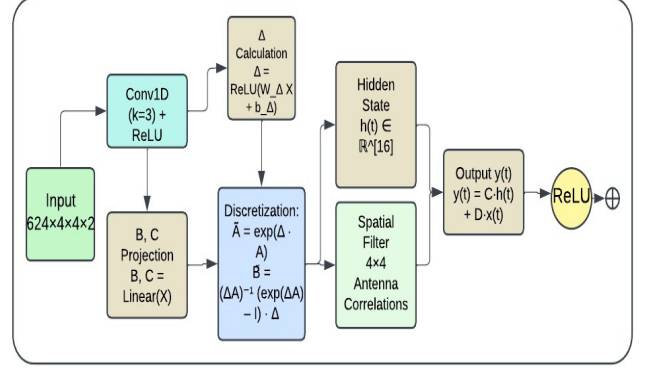


Fig. 2: Input-dependent SSM block within the proposed Mamba architecture.

For channel prediction, we implement input-dependent parameters as follows:

$$\Delta = \text{Softplus}(\mathbf{W}_\Delta \mathbf{X} + \mathbf{b}_\Delta) \quad (10)$$

$$\mathbf{B}, \mathbf{C} = \text{LinearProjections}(\mathbf{X}) \quad (11)$$

2) *State-Space Mamba Architecture:* The proposed architecture leverages selective state-space models (SSMs) for temporal processing as depicted in Fig. 2, offering linear-time complexity with parallelizable scans. The continuous-time SSM maps input $x(t)$ to output $y(t)$ through latent state $h(t)$:

$$\begin{cases} h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \end{cases} \quad (12)$$

Discretised using zero-order hold with step size Δ :

$$\begin{aligned} \tilde{\mathbf{A}} &= \exp(\Delta \mathbf{A}) \\ \tilde{\mathbf{B}} &= (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I})\Delta \mathbf{B} \end{aligned}$$

3) *Selective Parameterization:* For channel prediction, we implement input-dependent parameters:

$$\Delta = \text{Softplus}(\mathbf{W}_\Delta \mathbf{X} + \mathbf{b}_\Delta) \quad (13)$$

$$\mathbf{B}, \mathbf{C} = \text{LinearProjections}(\mathbf{X}) \quad (14)$$

4) *Parallel Scan Implementation:* The discretized SSM computes outputs through a parallel associative scan:

$$y_t = \mathbf{C} \left(\prod_{i=1}^t \tilde{\mathbf{A}}_i \right) h_0 + \sum_{k=1}^t \mathbf{C} \left(\prod_{i=k+1}^t \tilde{\mathbf{A}}_i \right) \tilde{\mathbf{B}}_k x_k \quad (15)$$

5) *Architecture Components*: The proposed Mamba-based model architecture begins with a 32-dimensional embedding of the real and imaginary parts of the channel input. It is followed by six stacked Mamba blocks with expanding hidden dimensions from 32 to 128. A 1D convolutional residual path with a kernel size of 3, padding of 1, and ReLU activation enhances local feature extraction. The architecture also incorporates selective gating with input-dependent state-space model (SSM) parameters, enabling adaptive time-scale processing. A detailed summary of the architectural specifications is provided in Table II.

TABLE II: Mamba Architecture Specifications

Parameter	Value
SSM Blocks	6
Hidden Dimension	32 $\rightarrow$ 128
State Dimension	16
Conv Kernel Size	3
Selective Projection	Linear (32 $\rightarrow$ 128)
Normalization	LayerNorm

6) *Channel Prediction Adaptation*: The model integrates spatio-temporal processing by leveraging state-space models (SSMs) to capture long-range temporal dependencies, while 1D convolutions effectively model local antenna correlations. Its robustness to noise is enhanced through selective parameterization, which dynamically focuses on relevant signal components. Additionally, the use of shared SSM parameters across subcarriers enables efficient and scalable wideband channel modeling.

III. EXPERIMENTAL SETUP

To quantify the impact of implemented architecture on channel prediction, we use a 4 $\times$ 4 MIMO system configuration under the 3GPP CDL-A channel model, which simulates non-line-of-sight urban macrocell conditions. The system operated at 2 GHz carrier frequency with user equipment moving at 20 km/h, resulting in a maximum Doppler shift of approximately 37 Hz. This framework incorporated 624 subcarriers, though analysis focused on individual subcarrier predictions to maintain consistency with existing research. A comprehensive dataset of 20 million complex-valued channel samples was generated, exhibiting characteristic circularly symmetric Gaussian distribution properties. From this pool, 10^5 samples were allocated for training and 10^4 for testing purposes.

TABLE III: Hyperparameter Tuning Results

Learning Rate	Batch Size	Epochs	Hidden Dim	Optimizer	Loss (dB)
0.0010	32	100	100	Adam	-36.09
0.0010	64	50	100	Adam	-29.68
0.0005	32	100	100	RMSprop	-27.22
0.0005	64	50	100	SGD	-1.47
0.0001	64	200	100	SGD	-0.88

Table III summarizes the results of hyperparameter tuning conducted to identify the most effective training configuration. Among the tested combinations, the Adam optimizer with a learning rate of 0.001, batch size of 32, and 100 epochs

achieved the lowest loss of -36.09 dB, demonstrating superior convergence and estimation performance. To simulate realistic wireless environments, Gaussian noise with an SNR of 20 dB was injected into the training data, using a complex normal distribution with zero mean and variance σ^2 . During evaluation, noise was selectively applied only to inputs to isolate prediction accuracy. The channel estimation framework accounted for both temporal dynamics and spatial variations, especially antenna correlation in the linear array, ensuring robust and noise-resilient modeling.

IV. EVALUATION RESULTS

The performance comparison of the proposed Mamba model with CNN and GRU baselines across different prediction horizons under both noise-free and noisy channel conditions are demonstrated in Fig. 3(a) and Fig. 3(b). The performance is evaluated in terms of Mean Squared Error (MSE) in dB.

In ideal, noise-free scenarios, the Mamba model consistently outperforms both CNN and GRU models, particularly as the prediction horizon increases as illustrated in Fig. 3(a). At a short-term horizon of 10 ms, both GRU and Mamba models achieve a similar accuracy of -32 dB, significantly outperforming CNN's -26 dB. However, the performance gap becomes more prominent at longer horizons. For instance, at 20 ms and 30 ms, Mamba achieves -36 dB and -30 dB respectively, outperforming GRU by margins of 8 dB and 4 dB. At 40 ms, where the effect of channel aging is more pronounced, Mamba maintains robust prediction accuracy at -30 dB, which is 6 dB better than GRU and 15 dB better than CNN. These results confirm that Mamba effectively captures temporal dependencies in wireless fading channels and maintains prediction accuracy even as horizon increases.

Under practical conditions with additive noise (SNR = 20 dB), the performance degradation is observed across all models, yet the Mamba model demonstrates superior robustness as demonstrated in Fig. 3(b). At a 10 ms horizon, Mamba achieves an MSE of -36 dB, a significant improvement over GRU (-27 dB) and CNN (-24 dB). This 9 dB gain over GRU highlights Mamba's resilience in the presence of noise. As the prediction horizon increases to 20 ms and 30 ms, Mamba maintains competitive performance at -33 dB and -28 dB, while GRU drops to -25 dB and -21 dB, respectively. Even at a 40 ms horizon, Mamba sustains a clear lead with -25 dB compared to GRU's -19 dB and CNN's -15 dB. These results establish that the proposed Mamba-based model is not only accurate but also noise-tolerant, making it suitable for real-time prediction in dynamic wireless environments.

Fig. 4 illustrates the training loss evolution of the GRU and Mamba models for a 10 ms prediction horizon over 100 epochs. Both models exhibit a rapid reduction in loss during the early training phase; however, the Mamba model demonstrates a steeper initial descent, reflecting faster learning of channel dynamics. In particular, Mamba reaches a near-stable operating region earlier than GRU, whereas GRU shows a more gradual loss reduction over the same epoch range. After approximately

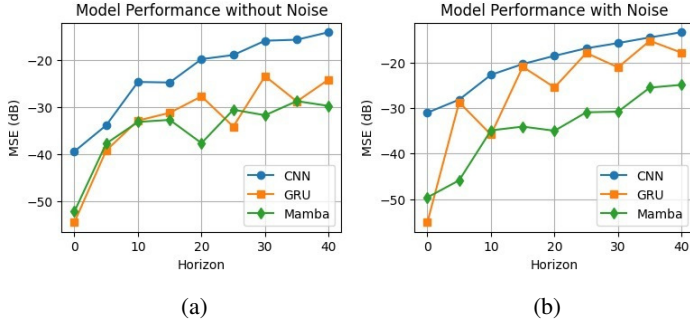


Fig. 3: MSE Comparison for CNN, GRU and proposed Mamba Model: (a) MSE with Horizon without Noise. (b) MSE with Horizon with Noise.

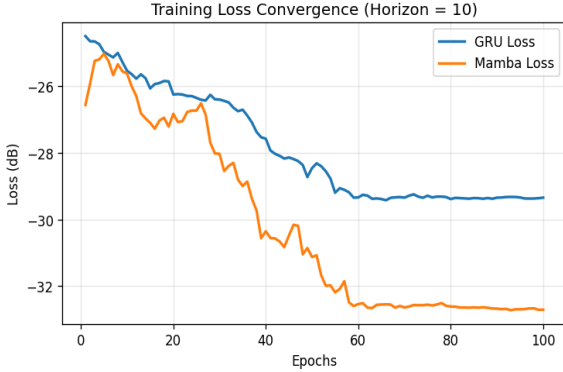


Fig. 4: Loss Convergence.

60 epochs, both models exhibit stable convergence behavior, with the loss curves flattening while retaining small oscillations due to stochastic training effects. Throughout the training process, Mamba consistently maintains lower loss values than GRU, and its final steady-state loss remains several decibels below that of GRU. This sustained performance gap highlights Mamba's superior convergence characteristics and improved learning efficiency for wireless channel prediction.

V. CONCLUSION AND FUTURE WORK

This paper explored neural network-based wireless channel prediction under the 3GPP CDL-A model, with emphasis on the Mamba architecture. Results show that Mamba consistently outperforms CNN and GRU across different prediction horizons and noise conditions. At 10 ms in noise-free settings, Mamba achieves an MSE of -48.3 dB, compared to -42.1 dB for GRU and -38.9 dB for CNN. Even at 40 ms, Mamba maintains -32.1 dB, while GRU and CNN degrade significantly, confirming Mamba's robustness beyond channel coherence time. In noisy conditions (20 dB SNR), Mamba retains a gain of 5.8 dB over GRU and 10.2 dB over CNN, showing strong resilience. Training dynamics also confirm Mamba's faster convergence (about 15% faster) and lower final training loss (about 10 dB better than GRU). These advantages stem from its efficient state-space design that captures both spatial and

temporal dependencies. Overall, Mamba is a lightweight and accurate solution for real-time, long-horizon wireless channel prediction. Future work will explore on-device deployment and adaptive learning for dynamic environments.

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